

[Sleep apnea syndrome and obesity].

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Obesity is a main risk factor for sleep apnea syndrome (SAS). The prevalence of SAS is especially high in massive obesity and in visceral obesity. The mechanisms of obstructive apneas in obesity are poorly known, but an increase in upper airway collapsibility probably plays an important role. Several cardiorespiratory complications of SAS, especially systemic arterial hypertension, diurnal alveolar hypoventilation and pulmonary arterial hypertension, are more frequent and more severe in obese patients. An important weight loss resulting from surgical treatment of obesity is often associated with a dramatic decrease in apnea-hypopnea index in patients with massive obesity. In patients with moderate obesity, dietary weight loss is associated with varying degrees of improvement in SAS. Pharyngoplasty and anterior mandibular positioning devices have a low success rate in patients with massive obesity. Nasal continuous positive airway pressure is often the only effective treatment in obese SAS patients.

[Obesity and obstructive sleep apnea in children].

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Obesity, along with hypertrophy of the adenoids and the tonsils, represents one of the major risk factors for obstructive sleep apnea (OSA) in children. Obesity is associated with an increase in the prevalence and the severity of OSA and is a major factor in the persistence and aggravation of OSA over time. Neurocognitive dysfunction and abnormal behavior are the most important and frequent end-organ morbidities associated with OSA in children. Other deleterious consequences such as cardiovascular stress and metabolic syndrome are less common in children than in adults with OSA. Defining the exact role of obesity in OSA-associated end-organ morbidity in children is difficult because of the complex and multidimensional interactions between sleep in general, OSA, obesity, and metabolic dysregulation. This may explain why obesity itself has not been shown to be associated with a significant increase in OSA-associated end-organ morbidity. Obesity is linked to a decreased treatment efficacy and, in particular, of adenotonsillectomy. Peri- and postoperative complications are more common and more severe in obese children as compared with normal-weight controls. Continuous positive airway pressure (CPAP) is frequently needed, but compliance with CPAP is less optimal in obese children than in non-obese children. In conclusion, obesity represents a major public health problem worldwide; its prevention is one of the most efficient tools for decreasing the incidence and the morbidity associated with OSA in children.

Obesity and obstructive sleep apnea: or is it OSA and obesity?

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Obstructive sleep apnea (OSA) consists of repetitive choking spells due to sleep-induced reduction of upper airway muscle tone. Millions of adults and children live unaware of this condition, which can have a profound affect on their health and quality of life. Obesity, gender, genetic, and hormonal factors mediate risk for OSA and interact in a multifaceted manner in the pathogenesis of this disease. Obesity is the most established and primary risk factor given that body mass index, visceral fat, and neck circumference are major predictors in the clinical expression of OSA. Many studies have shown weight

loss or gain significantly impacts OSA severity. More recently, accumulating evidence indicates OSA promotes weight gain, obesity, and type II diabetes in a variety of ways, such that obesity and OSA form multiple interleaved vicious cycles. Thus, creative strategies to increase physical activity, improve diet, and otherwise facilitate weight management become particularly vital given the epidemics of obesity and OSA in the United States. In this regard, the American College of Sports Medicine recently launched the "Exercise is Medicine" (initiative exerciseismedicine.org). In the future, medications may emerge to treat obesity, OSA, and their sequelae with minimal side effects. However, there are effective ways to approach these problems now without waiting for "the magic pill".

Obesity, ethnicity, and sleep-disordered breathing: medical and health policy implications.

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The obesity epidemic in the United States is a major health problem, which poses immediate and long-term health risks with potential for significant reduction in life expectancy if not curtailed. Despite affecting all segments of the population, minorities carry a relatively larger burden of the obesity threat. Obesity is a major risk factor for developing obstructive sleep apnea and the metabolic syndrome, which are associated with increased cardiovascular and cerebrovascular morbidity and mortality. This article discusses the implications of the obesity trend on the prevalence of obstructive sleep apnea and health, highlights ethnic differences, reviews the resultant economic implications, and suggests public health and health policy initiatives that may be beneficial in stemming the obesity scourge.

The reciprocal interaction between obesity and obstructive sleep apnoea.

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Obesity is a significant risk factor in the pathogenesis of obstructive sleep apnoea (OSA) altering airway anatomy and collapsibility, and respiratory control. The association between obesity and OSA has led to an increasing focus on the role of weight loss as a potential treatment for OSA. To date, most discussion of obesity and OSA assumes a one-way cause and effect relationship, with obesity contributing to the pathogenesis of OSA. However, OSA itself may contribute to the development of obesity. OSA has a potential role in the development and reinforcement of obesity via changes to energy expenditure during sleep and wake periods, dietary habits, the neurohormonal mechanisms that control satiety and hunger, and sleep duration arising from fragmented sleep. Thus, there is emerging evidence that OSA itself feeds back into a complex mechanism that leads either to the development or reinforcement of the obese state. Whilst current evidence does not confirm that treatment of OSA directly influences weight loss, it does suggest that the potential role OSA plays in obesity and weight loss deserves further research.

Approach the Patient With Obstructive Sleep Apnea and Obesity.

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Obstructive sleep apnea (OSA) and obesity are highly prevalent and bidirectionally associated. OSA is

underrecognized, however, particularly in women. By mechanisms that overlap with those of obesity, OSA increases the risk of developing, or having poor outcomes from, comorbid chronic disorders and impairs quality of life. Using 2 illustrative cases, we discuss the relationships between OSA and obesity with type 2 diabetes, dyslipidemia, cardiovascular disease, cognitive disturbance, mood disorders, lower urinary tract symptoms, sexual function, and reproductive disorders. The differences in OSA between men and women, the phenotypic variability of OSA, and comorbid sleep disorders are highlighted. When the probability of OSA is high due to consistent symptoms, comorbidities, or both, a diagnostic sleep study is advisable. Continuous positive airway pressure or mandibular advancement splints improve symptoms. Benefits for comorbidities are variable depending on nightly duration of use. By contrast, weight loss and optimization of lifestyle behaviors are consistently beneficial.

Obstructive sleep apnea is associated with increased arterial stiffness in severe obesity.

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Obstructive sleep apnea is associated with obesity and metabolic syndrome, leading to greater cardiovascular risk. Severely obese patients with obstructive sleep apnea may still be at risk of adverse health outcomes, even without previous cardiovascular disease. Pulse wave analysis non-invasively measures peripheral pulse waveforms and derives measures of haemodynamic status, including arterial stiffness, augmentation pressure and subendocardial viability ratio. We hypothesized that the presence of obstructive sleep apnea in severe obesity, even in the absence of an antecedent history of cardiovascular disease, would affect measurements derived from pulse wave analysis. Seventy-two severely obese adult subjects [obstructive sleep apnea 47 (body mass index 42 ± 7 kg m⁻²), without obstructive sleep apnea (non-OSA) 25 (body mass index 40 ± 5 kg m⁻²)] were characterised using anthropometric, respiratory and cardio-metabolic parameters. Groups were similar in age, body mass index and gender. More subjects with obstructive sleep apnea had metabolic syndrome [obstructive sleep apnea 60%, without obstructive sleep apnea (non-OSA) 12%]. Those with obstructive sleep apnea had greater arterial stiffness, augmentation pressure and decreased subendocardial viability ratio (all $P < 0.001$), with significantly higher systolic ($P = 0.003$), diastolic ($P = 0.04$) and mean arterial pressures ($P = 0.004$) than patients without obstructive sleep apnea (non-OSA). Arterial stiffness correlated with mean arterial blood pressure ($P = 0.003$) and obstructive sleep apnea severity (apnea-hypopnea index; $P < 0.001$). apnea-hypopnea index significantly predicted arterial stiffness in multiple regression analysis, but components of the metabolic syndrome did not. Thus, patients with obstructive sleep apnea with severe obesity have increased arterial stiffness that may potentially influence cardiovascular risk independently of metabolic abnormalities. The presence of obstructive sleep apnea in severe obesity identifies a group at high cardiovascular risk; clinicians should ensure that risk factors are managed appropriately in this group whether or not treatment of obstructive sleep apnea is offered or accepted by patients.

Evaluation of obesity and asthma as risk factors for moderate to severe obstructive sleep apnea in children.

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Asthma and obesity are risk factors for obstructive sleep apnea (OSA) in children but their link to OSA

severity is uncertain. We aimed at determining whether asthma or obesity was associated with an increased risk of moderate/severe OSA. Children undergoing a one-night polysomnography for suspicion of OSA were retrospectively included. Univariate and multivariate analyses were conducted to assess the clinical and demographic characteristics linked to moderate/severe OSA (obstructive apnea-hypopnea index ≥ 5 events/h of sleep) with odds ratio and 95% confidence interval reported. Four hundred ninety children (311 [64%] boys) were included with a median [25th; 75th percentile] age of 8.7 [5.4; 12.9] years, 164 (33%) nonasthmatics nonobese, 122 (25%) obese nonasthmatics, 125 (26%) asthmatics nonobese, 79 (16%) asthmatics and obese. Moderate/severe OSA was present in 157 (32%) children (75/157 [48%] obese and 52/157 [33%] asthmatics). Independent factors associated with increased or decreased risk of moderate/severe OSA were obesity and male sex (odds ratio 1.82 [1.16; 2.87], $P = .01$, and 1.55 [1.02; 2.36], $P = .04$, respectively), and current asthma, age > 6 years, or behavioral disorders (odds ratio 0.45 [0.29; 0.70], $P < .001$; 0.44 [0.27; 0.73], $P < .001$; and 0.55 [0.33; 0.92], $P = .02$, respectively). Abnormal resistance of the respiratory system (measured in 241 children), but not abnormal spirometry (measured in 213 children), increased the risk of moderate/severe OSA (odds ratio 2.95 [1.46-5.96], $P = .003$). In our cohort enriched in obese and asthmatic children, obesity was associated with higher risk of moderate/severe OSA whereas current asthma was not. Ioan I, Mulier G, Taytard J, C  saire A, Beydon N. Evaluation of obesity and asthma as risk factors for moderate to severe obstructive sleep apnea in children. *J Clin Sleep Med*. 2022;18(6):1639-1648.

Obesity in children with different risk factors for obstructive sleep apnea: a community-based study.

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This study investigated the association between obesity and obstructive sleep apnea (OSA) in preschool and school-age children. Parents of obese and randomly chosen normal weight children completed a questionnaire on sleep-related symptoms, demography, family, and medical history. All subjects were invited to undergo polysomnography (PSG). OSA cases were defined as obstructive apnea hypopnea index (OAHI) ≥ 1 . A total of 5930 children were studied with 9.5% obese (11.9% boys/6.1% girls), 205/2680 preschool and 360/3250 school children. There were 1030 children (535 obese/495 normal weight) who underwent PSG. OSA was higher in obese children and obese school children had higher OAHI, arousal index, and shorter total sleep time. However, there was no positive correlation between OSA and body mass index (BMI). The main risk factors for OSA in preschool children were adenotonsillar hypertrophy and recurrent respiratory tract infection. The main cause for OSA in school children was a history of parental snoring and obesity. Mallampati scores and sleep-related symptoms were found to be associated with OSA in both preschool and school children. We demonstrated differential risk factors for OSA in obese children, which suggest that a different mechanism may be involved in OSA development in preschool and school-age children. Various risk factors have been reported in obese children with OSA owing to the different age and different study design. Obese children have a higher prevalence and severity of obstructive sleep apnea (OSA). OSA risk factors in obese children are affected by different ages and study designs. A differential prevalence and risk factors for obese preschool and school-age children with OSA has been demonstrated.

[How do you really know if the obese patient has sleep apnea?].

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Obesity is known to predispose to obstructive sleep apnea (OSA), a condition characterized by repeated episodes of apnea or hypopnea during sleep, due to the interruption of airflow through the nose and mouth. These episodes lead to the fragmentation of sleep and to decrease in oxyhaemoglobin saturation. Patients with massive obesity, with or without daytime hypersomnolence should be systematically screened for OSA, because many of them appear to be asymptomatic and unaware of their breathing abnormalities during sleep. Polysomnography (PSG) in an attended hospital laboratory setting is the gold standard for the diagnosis of OSA. However portable recording devices can be used for screening with good sensibility and specificity, and even for diagnosis when the apnea-hypopnea index is high. However the final diagnosis can only be carried out in a sleep laboratory using PSG by highly-qualified personnel, because of the limitations of the portable recording device. There is a strong association between OSA and the risk of traffic accidents. It has been established that OSA affects quality of life. There is also increasing evidence that OSA is an independent risk factor for cardio-vascular diseases. This has been successfully demonstrated for hypertension by prospective studies. But the evidence remains weak for myocardial infarction, stroke or mortality. Treating OSA with continuous positive airway pressure (CPAP) is the treatment of choice. CPAP improves quality of life, driving simulator performance, blood pressure and sleepiness, as demonstrated by randomised placebo controlled trials. The majority of obese OSA patients are currently not being offered diagnosis testing and treatment. It's a real challenge due to the epidemic increase of obesity prevalence. Portable recording devices could be available outside the sleep laboratory in nutrition department, where morbid obesity is treated. This emphasizes the need for a real collaboration between these departments and sleep experts.